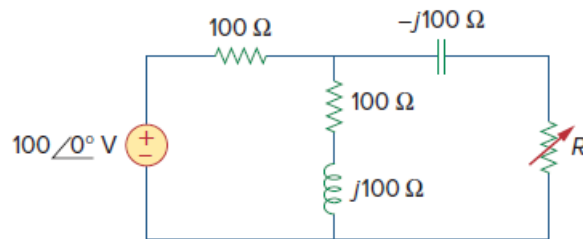


Problem

The variable resistor R in the circuit of Fig. 11.50 is adjusted until it absorbs the maximum average power. Find R and the maximum average power absorbed.

Figure 11.50:

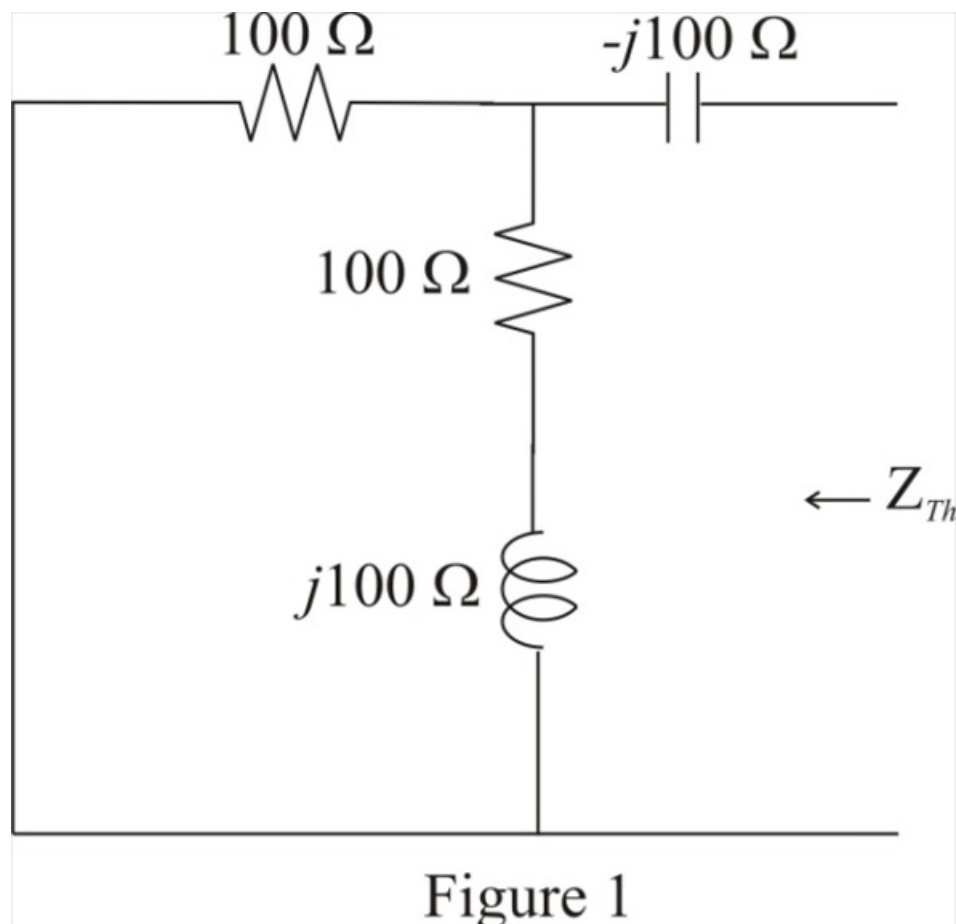


Step-by-step solution

Step 1 of 8

Refer to Figure 11.50 in the text book.

Deactivate the voltage source and disconnect the load resistance. Consider the circuit diagram shown in Figure 1.



Step 2 of 8

Calculate the value of the Thevenin's impedance Z_{Th} .

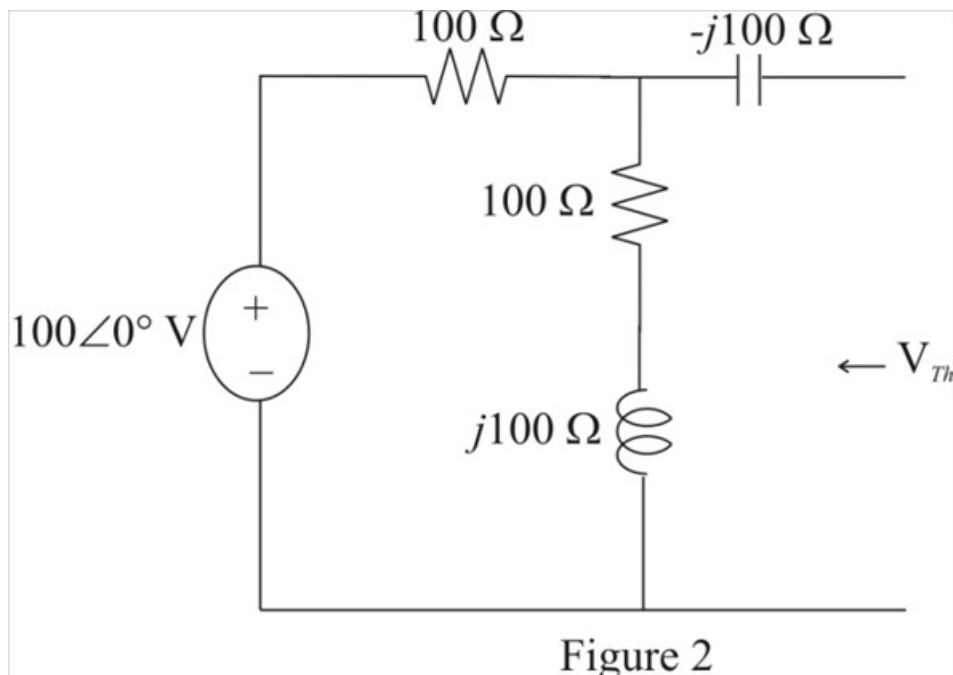
$$\begin{aligned} Z_{Th} &= [100 \parallel (100 + j100)] - j100 \\ &= \frac{(100 + j100)(100)}{100 + j100 + 100} - j100 \\ &= \frac{(100 + j100)(100)}{200 + j100} - j100 \\ &= \frac{100 + j100}{2 + j1} - j100 \end{aligned}$$

$$\begin{aligned} Z_{Th} &= 60 + j20 - j100 \\ &= 60 - j80 \, \Omega \end{aligned}$$

Therefore, the value of the Thevenin's impedance Z_{Th} is, $60 - j80 \, \Omega$.

Step 3 of 8

Consider the circuit diagram shown in Figure 2 to calculate the value of the Thevenin's voltage.



Step 4 of 8

Calculate the value of the Thevenin's voltage V_{Th} using voltage division rule.

$$\begin{aligned} V_{Th} &= (100\angle 0^\circ) \left(\frac{100 + j100}{100 + 100 + j100} \right) \\ &= (100\angle 0^\circ) \left(\frac{100 + j100}{200 + j100} \right) \\ &= (100\angle 0^\circ) (0.633\angle 18.44^\circ) \\ &= 63.3\angle 18.44^\circ \text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, $63.3\angle 18.44^\circ \text{ V}$.

Step 5 of 8

Calculate the value of the load resistance R_L that will absorb the maximum power.

$$R_L = |Z_{Th}|$$

Substitute $60 - j80 \Omega$ for Z_{Th} .

$$\begin{aligned} R_L &= |60 - j80| \\ &= \sqrt{(60)^2 + (80)^2} \\ &= \sqrt{3600 + 6400} \\ &= \sqrt{10000} \end{aligned}$$

$$R_L = 100 \Omega$$

Therefore, the value of the load resistance R_L that will absorb the maximum power is,

$$\boxed{100 \Omega}.$$

Step 6 of 8

Draw the Thevenin's equivalent circuit diagram.

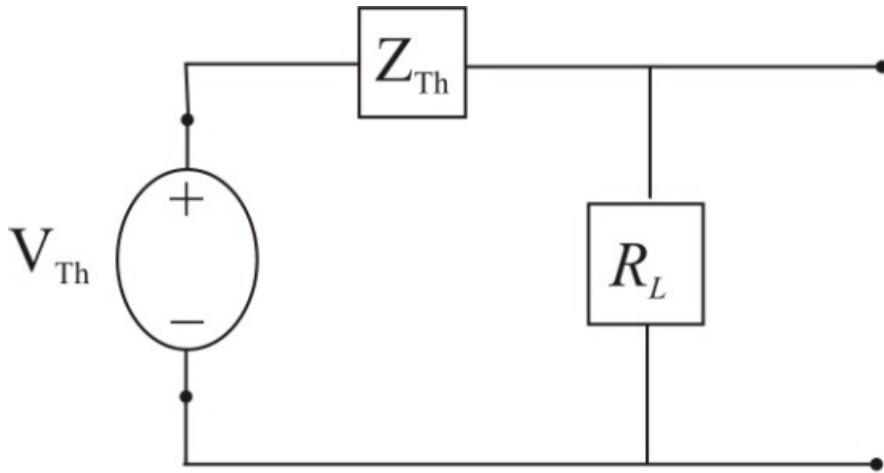


Figure 3

Step 7 of 8

Calculate the value of the current I flowing in the circuit.

$$I = \frac{V_{Th}}{Z_{Th} + R_L}$$

Substitute $63.3\angle 18.44^\circ \text{ V}$ for V_{Th} , $60 - j80 \Omega$ for Z_{Th} and 100Ω for R_L .

$$\begin{aligned} I &= \frac{63.3\angle 18.44^\circ}{60 - j80 + 100} \\ &= \frac{63.3\angle 18.44^\circ}{160 - j80} \\ &= \frac{63.3\angle 18.44^\circ}{178.889\angle -26.57^\circ} \\ &= 0.354\angle 45.01^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I flowing in the circuit is, $0.354\angle 45.01^\circ \text{ A}$.

Step 8 of 8

Calculate the value of the maximum average power $P_{\max}$ absorbed by the load resistance.

$$P_{\max} = \frac{1}{2} |I|^2 R_L$$

Substitute 0.354 A for $|I|$ and $100 \, \Omega$ for R_L .

$$\begin{aligned} P_{\max} &= \frac{1}{2} (0.354)^2 (100) \\ &= (0.1253)(50) \\ &= 6.265 \text{ W} \end{aligned}$$

Therefore, the value of the maximum average power $P_{\max}$ absorbed by the load resistance is,

6.265 W.